

LESSON 103 (2.0) GROUND

Lesson Objective

During this lesson, the applicant will be introduced to normal checklist procedures and how the checklist is used. The applicant will also become familiar with the flight deck layout and conduct a preflight inspection with the instructor.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Ask the student if they got stuck at any point in their practice weight and balance problems.
- Student questions.

References

- Advisory Circulars. ([Search Tool](#))
 - Pilot's Guide to a Preflight Briefing. (AC 91-92)
 - Aeronautical Decision Making. (AC 60-22)
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 07 - Aircraft Systems.
- Warrior III Pilot's Information Manual.
 - Section 01 - General.
 - Section 07 - Systems.

Instructor Guidance

- Encourage your student to refer to the amplified checklist to review any items they are unclear on.
- Have your student take photos of the flight deck for later review.
- Emphasize that checklists are a non-negotiable safety habit, not a formality — instill this mindset early, as it will carry through their entire aviation career.
- When conducting the preflight inspection, narrate your actions aloud so the student can follow your thought process, then have them repeat the inspection themselves.
- Point out any airworthiness discrepancies you find — or commonly find — during the preflight as a teaching opportunity. Even minor items help build the student's eye for a thorough inspection and reinforce why the preflight matters.
- When introducing the engine controls, have the student physically touch and operate the throttle, mixture, fuel selector, and primer while the engine is off. Familiarity with the feel and location of each control before the first flight reduces cognitive load in the cockpit significantly.

Lesson Content

Checklist Procedures

- Emphasize the importance of proper checklist usage and its impact on safety.
- Show location of checklists.
 - Checklists must be bought at customer service! They are also available digitally on the Florida Tech Hub.
 - If your student has an Ipad, you could direct them to FTA app as well.
- Distinguish between normal and emergency checklists.
- Emphasize the difference between read-and-do checklists and do-and-verify checklists.
- Student pilots in stage 1 are permitted to use the booklets as substitute for the one-page checklist if they wish.

Flight Deck Familiarization

- Take the student out to the airplane and have them sit in the flight deck.
- Be sure to point out that switches are not always in the same location depending on the airplane.
- Locate and describe basic flight deck elements and controls.

Preflight Inspection

- Emphasize the importance of a proper pre-flight inspection.
- Checklists **MUST** be used while pre-flighting.
- Conduct a pre-flight inspection with the student.
- Introduce AC 91-92 to the student.

Operation of Engine Controls

- Throttle control
 - Regulates engine power output by controlling the amount of air-fuel mixture entering the cylinders.
 - Pushing the throttle increases power (RPM) for climb or takeoff.
 - Reducing throttle decreases power for descent or landing.
 - Smooth adjustments prevent engine stress or abrupt attitude changes.
- Mixture control
 - At sea level, mixture is full rich for maximum power and cooling.
 - As altitude increases, air density decreases, so the mixture must be leaned for efficient combustion.
 - Overly rich → rough running, loss of power, carbon buildup.
 - Overly lean → high temperatures, detonation, or engine damage.

- Fuel Selector
 - Chooses which tank(s) the engine draws fuel from (Left, Right, Both).
 - Fuel tanks must be switched every 30min
- Primer
 - Adds small amounts of fuel directly to the cylinders before starting a cold engine.
 - Always verify fuel selector position during preflight and in emergencies.
 - Excessive priming can cause flooding.

Post-Brief

- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 104.
- Assign homework:
 - Review the checklists and begin to memorize all memory items and emergency checklists.
 - Prepare a TOLD card for the 104 sim using an airplane assigned by the instructor.
 - Read chapter 03 of the Airplane Flying Handbook - Basic Flight Maneuvers:
 - The four fundamentals.
 - Effects and use of flight controls.
 - Attitude flying.
 - Straight and level flight.

Completion Standards

This lesson is complete when the applicant can:

1. Understand checklist procedures to include all elements on the preflight inspection.
2. Locate and describe basic flight deck elements and controls.
3. Operate the engine controls in a safe manner.